Genre in Grammar: PCA on Sanskrit Morphological Variation

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Abstract

Sanskrit literary tradition spans over 2,400 years and encompasses five major genre categories—Vedic, Epic, Kavya, Shastra, and Puranic—whose morphological distinctiveness has never been quantitatively established at scale. In this project we perform large-scale computational analysis of genre-conditioned morphological variation using the Digital Corpus of Sanskrit (DCS), comprising 5.69 million tokens across 270 texts. We apply Principal Components Analysis (PCA) to a 30-dimensional feature vector of normalized morphological and syntactic frequencies per text, revealing a dominant *nominalization axis* (PC1, 30.3% variance) that separates verbal-heavy Vedic texts from nominal-heavy Shastra and Kavya. Supervised classification with class-weight-balanced Logistic Regression achieves 63.2% macro F1 while unsupervised k-means clustering (ARI $= 0.21$) reveals that only Vedic forms a morphologically coherent cluster. Spearman correlations between text period and morphological features ($r = -0.66$ for MCI, $p < 0.0001$) confirm a systematic diachronic shift from morphological complexity toward compounding across Sanskrit's attested history. Code is available on: https://github.com/gargishroff/Genre-in-Grammar.git

1 Introduction

Sanskrit is one of the most morphologically rich and diachronically documented languages in the world, with a continuous textual tradition spanning from the Rigveda (∼1100 BCE) to early modern compositions (∼1900 CE). Its literary output is organized into five broadly recognized genre categories: Vedic (Samhitas, Brahmanas, Aranyakas, Upanishads, and ritual sutras), Epic (Mahabharata, Ramayana), Kavya (classical poetry and drama), Shastra (technical scientific prose covering grammar, medicine, law, philosophy, and alchemy), and Puranic (sectarian encyclopedic narratives)(Warder, 1972–2011; Keith, 1920). These genres differ dramatically in register, style, and compositional period—yet whether these differences manifest systematically in *morphological* feature distributions has never been tested at scale, leaving a fundamental question about the relationship between literary tradition and grammatical structure unanswered.

The Digital Corpus of Sanskrit (DCS)(Hellwig, 2010–2021) provides an unprecedented resource: over 5.4 million morphologically annotated tokens from 270 texts spanning the full range of Sanskrit literary history. This scale makes large-scale quantitative genre analysis feasible, enabling us to move from qualitative Indological description to computational verification of genre-morphology relationships.

We apply a fully computational pipeline to the DCS, extracting a 30-dimensional morphological feature vector per text (including POS tag rates, case distributions, tense and mood rates, compound rate, syntactic dependency features, and morphological complexity entropy), then applying PCA for dimensionality reduction, k-means and hierarchical clustering for unsupervised genre recovery, and three supervised classifiers (SVM, Logistic Regression, LDA) with class-weight balancing for genre classification. Spearman correlations quantify the diachronic dimension.

Our findings establish that Sanskrit genre variation is genuinely multidimensional (6 PCs required to explain 72% of variance), that Vedic is the only genre forming a coherent morphological cluster (100% k-means purity), that the Kavya–Shastra boundary is linguistically permeable in ways that reflect genuine prose-register overlap, and that Sanskrit morphological complexity declines monotonically across its 2,400-year documented history—providing the first large-scale quantitative evidence that Sanskrit genre is, at its core, a *Vedic versus Classical* morphological divide.

2 Related Work

2.1 Sanskrit NLP and morphological annotation

Morphological analysis of Sanskrit is computationally challenging due to the language's rich inflectional paradigm (8 cases, 3 numbers, 3 genders, complex verbal system) and pervasive sandhi (phonological fusion at word boundaries). The DCS(Hellwig, 2010–2021), built by Hellwig over fifteen years, provides the largest available morphologically annotated Sanskrit corpus, covering texts from 500 BCE to 1900 CE in CoNLL-U format with full case, tense, mood, number, gender, and compound annotations. Building on this infrastructure, Gupta et al. evaluated neural sequence labelling models for Sanskrit morphological tagging, demonstrating that transformer-based models trained on DCS annotations generalize across genres(Gupta et al., 2020)—an important prerequisite for the downstream genre classification task we undertake. These works establish that DCS annotations are reliable enough for computational genre analysis, grounding the validity of the 30-feature extraction pipeline we apply.

2.2 Genre classification and register variation in NLP

Automatic text genre and register classification has a rich history in computational linguistics. Manning and Schütze establish the theoretical foundations connecting distributional syntactic and morphological features to register distinction(Manning and Schütze, 1999). In the context of Indian language NLP, Kakwani et al. released the IndicNLPSuite, providing monolingual corpora and pretrained models for 11 Indian languages including Sanskrit(Kakwani et al., 2020), contextualizing our study within the broader effort to build computational resources for Indian languages. For Sanskrit specifically, genre classification has not previously been attempted at scale: existing work focuses on morphological tagging accuracy(Gupta et al., 2020) rather than genre-level variation. The present study is therefore the first to frame the DCS as a genre-labelled corpus and apply supervised and unsupervised methods to test whether the traditional literary taxonomy has a morphological basis.

2.3 Morphological complexity measurement

The operationalization of morphological complexity as Shannon entropy over morphological tag distributions—our Morphological Complexity Index (MCI)—follows directly from Bentz et al., who compare this and related complexity measures across typologically diverse corpora and validate their cross-linguistic interpretability(Bentz et al., 2016). Cotterell et al. provide a rigorous typological analysis of inflectional morphological systems, quantifying paradigm size and inflectional entropy across 98 languages; their framework positions Sanskrit as typologically extreme in morphological richness, particularly in its verbal system(Cotterell et al., 2018). De Haan surveys the typology of tense, aspect, and modality across languages, contextualizing the verbal mood distinctions (optative, subjunctive) that we find to be the primary Vedic–Classical discriminators in PC3(de Haan, 2009). Together, these works provide both the metric (MCI) and the typological frame for interpreting what it means for Sanskrit morphological complexity to decline over time.

2.4 PCA and dimensionality reduction in linguistic analysis

PCA is well-established as a tool for exploring continuous feature spaces in linguistics. Jolliffe and Cadima provide a comprehensive review of PCA methodology including the interpretation of loading vectors as feature contributions—directly relevant to our PC loading analysis(Jolliffe and Cadima, 2016). Hamilton et al. apply PCA to diachronic word embedding spaces to track semantic change across time, a methodological parallel to our diachronic morphological analysis(Hamilton et al., 2018). Our classification pipeline uses scikit-learn(Pedregosa et al., 2011) for SVM, Logistic Regression, and LDA with stratified cross-validation. These methodological precedents establish PCA as an interpretable and linguistically motivated tool for the feature spaces we explore.

2.5 Diachronic morphological change in Indo-European

The diachronic decline in morphological complexity documented here is consistent with well-established trends in Indo-European linguistics. Fortson surveys the historical development of Indo-European morphological systems, documenting the general tendency toward analytic structures as languages evolve from highly synthetic Proto-Indo-European(Fortson IV, 2010). Witzel documents the textual and social history of the Vedic canon, providing the chronological and genre-

level context within which our quantitative findings are grounded(Witzel, 1997). The transition from Vedic to Classical Sanskrit—marked by the Paninian grammatical standardization (Aṣṭādhyāyī, ∼400 BCE)—is one of the most precisely documented cases of deliberate morphological simplification in any language tradition, making Sanskrit an ideal case study for testing whether diachronic morphological change is detectable from corpus statistics alone.

3 Data and Methods

3.1 The Digital Corpus of Sanskrit

The DCS(Hellwig, 2010–2021) provides 5,688,416 tokens across 270 text folders spanning 500 BCE to 1900 CE, in CoNLL-U format with 10 columns per token: token ID, surface form, lemma, UPOS, DCS-specific XPOS, morphological features (FEATS), syntactic head (HEAD), dependency relation (DEPREL), enhanced dependencies, and a MISC field encoding `LemmaId`, `OccId`, `Unsandhied` form, `WordSem` sense IDs, and an `IsMantra` flag. Genre labels were assigned via a curated mapping of all 270 DCS text folder names (in IAST transliteration) to the five genre categories, achieving 270/270 coverage with no unassigned texts.

A critical parsing note: the FEATS value `Case=Cpd` flags compound members and does *not* represent a grammatical case(Hellwig, 2010–2021). All such tokens were flagged as `is_compound=True` and excluded from case frequency counts, ensuring clean case distribution features. The resulting corpus statistics (Table 1) reveal a severe class imbalance that motivates the balanced-weight classification strategy described in Section 4.5.

Table 1: DCS corpus statistics by genre. The imbalance between Shastra (126 texts, 46.7%) and Epic (4 texts, 1.5%) motivates the class-weight balancing strategy in classification.

Genre	Texts	Chapters	Tokens	Treebank chaps.
Epic	4	2,735	1,581,085	42
Vedic	76	7,343	1,316,026	1,692
Kavya	47	367	672,774	3
Shastra	126	4,308	1,380,603	25
Puranic	17	1,147	737,928	4
Total	270	15,900	5,688,416	1,766

3.2 Feature extraction

Each text is represented as a 30-dimensional feature vector, all rates normalized per 1,000 tokens:

- **POS rates** (8): NOUN, VERB, ADJ, ADP, PRON, NUM, ADV, PART
- **Nominal-to-verbal ratio**: (NOUN+ADJ+PRON)/max(VERB,1)
- **Case rates** (8): Nom, Acc, Inst, Dat, Abl, Gen, Loc, Voc (nominal tokens only)
- **Tense rates** (4): Pres, Past, Fut, Imp (verbal tokens only)
- **Mood rates** (4): Ind, Opt, Imp, Sub (verbal tokens only)
- **Compound rate**: tokens with `Case=Cpd` per 1,000
- **Mantra rate**: `IsMantra=True` tokens per 1,000
- **Syntactic features** (3): average dependency arc length, subordination rate (clausal deprels), verb-final rate—available only for the 1,766 Treebank chapters; zero elsewhere
- **MCI**: Shannon entropy (bits) over all morphological feature values per text(Bentz et al., 2016): $\mathrm{MCI} = -\sum_i p_i \log_2 p_i$

3.3 PCA pipeline

Features were standardized (zero mean, unit variance per feature) before PCA(Jolliffe and Cadima, 2016). The number of components retained (PC1–PC6, cumulative 72% variance) was determined by the scree plot elbow criterion. All downstream clustering and classification analyses use these six components exclusively.

3.4 Clustering

K-means ($k = 5$, 30 random restarts) and Ward hierarchical clustering were applied to PC1–PC6 scores. Cluster quality was evaluated with the Adjusted Rand Index (ARI) against true genre labels, where $\mathrm{ARI} = 1.0$ indicates perfect recovery and $\mathrm{ARI} = 0.0$ indicates chance-level assignment.

3.5 Classification with class balancing

The corpus is severely imbalanced: Shastra (126 texts, 46.7%) vs. Epic (4 texts, 1.5%). A classifier predicting Shastra for all texts would achieve 47% accuracy without learning anything about genre. To address this, we apply `class_weight='balanced'` to SVM (linear kernel) and Logistic Regression, and uniform priors to LDA. Each classifier trains on PC1–PC6 scores with 4-fold stratified cross-validation (capped by Epic's count of 4). Macro

F1—which equally weights all five genres regardless of support—is the primary evaluation metric, as overall accuracy is misleading given the class imbalance.

3.6 Diachronic analysis

Approximate BCE/CE dates were assigned to each text using a curated lookup table grounded in standard Sanskrit literary chronology(Warder, 1972–2011; Keith, 1920). Spearman rank correlations were computed between text period and each morphological feature, both globally and within each genre separately, to test whether the diachronic trend operates at the corpus level only or also within individual genres.

4 Results

4.1 PCA: dimensionality and principal axes

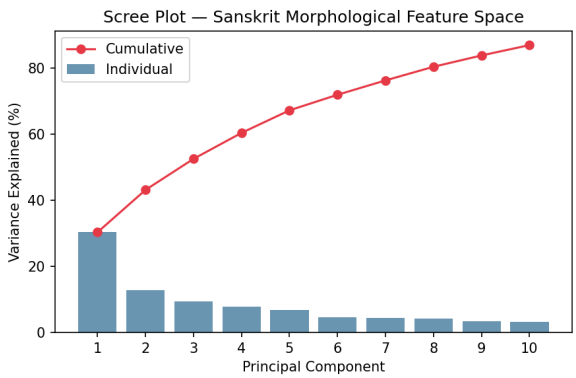

Figure 1: Scree plot of explained variance. PC1–PC6 cumulatively explain 72.0% of variance. The gradual elbow—rather than a sharp drop after PC1— indicates that Sanskrit genre variation is genuinely multidimensional rather than reducible to a single binary contrast.

Figure 1 shows that PC1 explains 30.3% of variance, with subsequent PCs contributing 12.8%, 9.4%, 7.9%, 6.8%, and 4.7% respectively. The gradual elbow indicates that Sanskrit genre variation is genuinely multidimensional: no single binary contrast accounts for the majority of morphological variance. This is a substantive finding in itself—it means that the five-genre taxonomy cannot be reduced to a Vedic-vs-Classical split alone, but requires at least six dimensions to adequately represent.

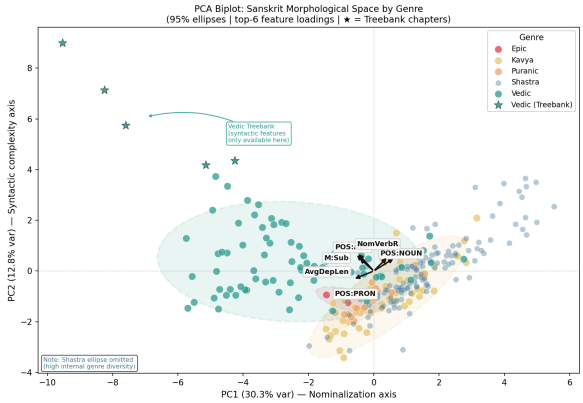

Figure 2: PCA biplot (PC1 vs. PC2). Vedic texts (teal) cluster far left on the nominalization axis; Shastra (blue) and Kavya (gold) cluster right. Stars (⋆) mark Vedic Treebank chapters. Shastra ellipse omitted due to high internal diversity.

Figure 3: Loading heatmap for the top 12 features across PC1–PC4. The optative mood loading on PC3 (+0.48) and verb-final rate on PC4 (+0.63) are the linguistically richest axes after PC1.

Figure 2 and the loading analysis (Figure 3) reveal four interpretable axes:

PC1 (nominalization axis): High-loading features are `pos_NOUN` (+0.32), `compound_rate` (+0.31), and `nom_verb_ratio` (+0.21); low-loading features are `pos_PRON` (−0.31), `mood_Sub` (−0.28), and `pos_VERB` (−0.27). Vedic texts score far left (verbal, pronominal, subjunctive-rich); Shastra and Kavya score right (nominal, compound-dense). This directly **confirms Hypothesis 1**: the nominalization axis is the dominant morphological axis separating Vedic from Classical Sanskrit

registers.

PC2 (syntactic complexity axis): `pos_ADP` (+0.36) and `avg_dep_length` (+0.27) drive PC2 upward; `pos_ADV` (−0.34) downward. The three extreme outliers at PC2>6 are Vedic Treebank chapters for which `avg_dep_length` is non-zero—a methodological artefact of partial syntactic annotation coverage(Hellwig, 2010–2021). PC2 does *not* primarily capture compound density as predicted; compound density co-varies with nominalization on PC1. **Hypothesis 2 is not confirmed**: compound density is a PC1 phenomenon, not an independent axis.

PC3 (archaic verbal mood axis): `mood_Opt` (+0.48) is the dominant loader, followed by `tense_Pres` (+0.35). The optative mood is the defining grammatical marker of archaic Vedic Sanskrit, virtually absent from Classical registers(de Haan, 2009). This axis captures the Vedic–Classical divide at a finer grammatical grain than PC1, distinguishing texts by their verbal mood inventory rather than their nominal density.

PC4 (word-order axis): `verb_final_rate` (+0.63) is the strongest single loading in the entire matrix, reflecting SOV verb-final tendency in Treebank chapters. Because this feature is only non-zero for the 1,766 annotated chapters, PC4 is primarily a Treebank-coverage axis and should be interpreted with this caveat.

Together, PC1–PC4 reveal that Sanskrit morphological space is structured along a Vedic–Classical nominalization gradient (PC1), an archaic mood inventory (PC3), and two artefact-influenced axes (PC2, PC4), establishing that genre differences are partially but not fully captured by morphological features alone.

4.2 Morphological Complexity Index by genre

Figure 4 confirms a clear MCI ranking: Vedic > Epic > Kavya ≈ Puranic > Shastra(Bentz et al., 2016; Cotterell et al., 2018). Vedic uses the full Sanskrit morphological inventory (8 cases, dual number, optative/subjunctive/injunctive moods, augmented past tense). The Shastra outliers below 2.5 bits—Nighaṇṭuśeṣa (1.87), Indu ad AHS (1.99), Pādārthacandrikā (2.03)—are vocabulary lists and philosophical definitional texts with severely restricted morphological range, where near-zero verbal inflection collapses the tag distribution entropy. Table 2 gives the full genre-level feature summary; the sevenfold difference in nominal-to-verbal ratio between Shastra (7.04) and Vedic (3.19) is the single most striking feature contrast in the corpus.

Table 2: Mean feature values by genre. The sevenfold difference in NVR between Shastra and Vedic, and the Vedic-exclusive mantra rate, are the most discriminating single features.

Genre	MCI	Cpd	Mantra	NOUN	NVR
Vedic	3.20	80.3	87.8	381.9	3.19
Epic	3.12	147.5	0.0	427.8	3.32
Kavya	3.03	241.4	0.0	444.0	4.01
Puranic	3.03	162.6	0.0	439.7	4.02
Shastra	2.84	236.5	0.0	502.5	7.04

NVR = nominal-to-verbal ratio; Cpd = compound rate (per 1,000)

4.3 Clustering

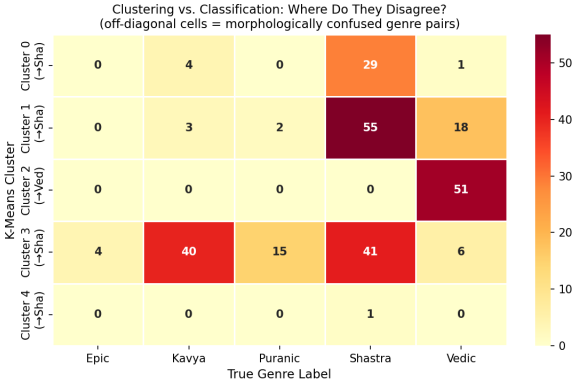

Figure 5: Clustering vs. classification heatmap. Cluster 2 recovers Vedic with 100% purity (51/51 texts). Cluster 3 co-locates 40/47 Kavya, 15/17 Puranic, and 41/126 Shastra texts—the primary site of morphological genre ambiguity in the corpus.

K-means achieved ARI = 0.210, indicating substantially below-chance genre recovery for non-Vedic genres. As shown in Figure 5, Cluster 2 recovers Vedic with perfect purity (51/51), directly **confirming Hypothesis 3**. The remaining four clusters are all Shastra-dominant, reflecting Shastra's numerical dominance (126/270 texts) and morphological overlap with other genres. Cluster 3 (39% pure) is the most revealing: it co-locates 40 Kavya, 15 Puranic, and 41 Shastra texts—demonstrating that these three genres share a morphological nominalization profile that unsupervised geometry cannot separate. Taken together, the clustering results establish that the traditional five-genre taxonomy does not correspond to five natural morphological clusters: only the Vedic–Classical divide is morphologically coherent, while the four Classical genres form a single overlapping region of morphological space.

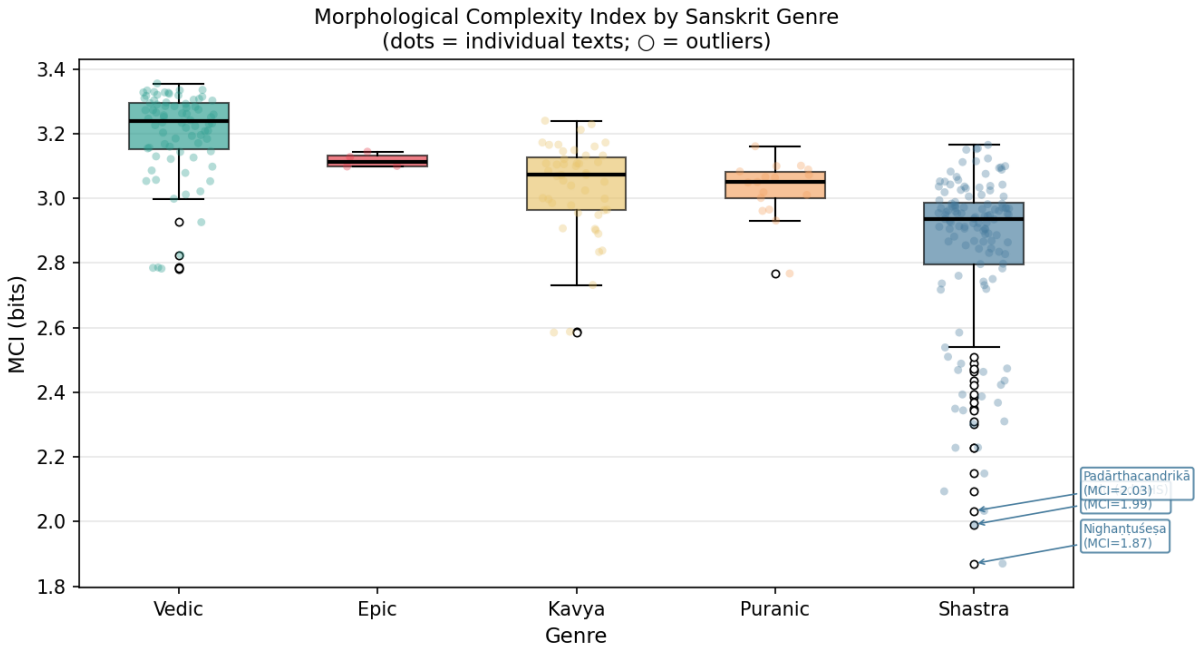

Figure 4: MCI by genre. Vedic has the highest median complexity (3.22 bits); Shastra the lowest (2.84 bits) with the widest spread. Annotated outliers Nighaṇṭuśeṣa (1.87 bits) and Pādārthacandrikā (2.03 bits) are a vocabulary list and a definitional commentary respectively. Dots show individual texts.

4.4 Classification benchmarking

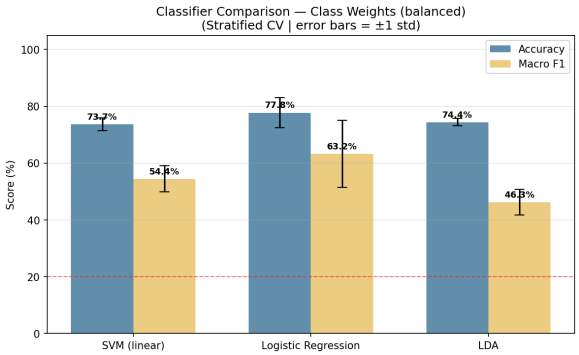

Figure 6: Classifier performance with balanced class weights. Logistic Regression achieves the highest macro F1 (63.2%), outperforming SVM (54.4%) and LDA (46.3%). The large accuracy–F1 gap for all classifiers reflects Shastra's class dominance inflating accuracy.

Table 3 presents the full per-genre classification results. Logistic Regression achieves the highest macro F1 (63.2%) with balanced class weights, **partially disconfirming Hypothesis 4** (which predicted SVM superiority). The accuracy–F1 gap for SVM (73.7% accuracy vs. 54.4% F1) reveals that high accuracy is partly an artefact of the 47% Shastra baseline: the weighted loss function forces the classifier to sacrifice some Shastra precision to recover minority-class recall.

Table 3: Per-genre recall with class-balanced classifiers (4-fold CV). LDA assigns zero recall to Puranic; Logistic Regression achieves the best overall minority-class performance.

Genre	SVM	LR	LDA
Epic	0.50	**0.75**	0.25
Kavya	0.55	**0.68**	0.53
Puranic	0.41	**0.41**	0.00
Shastra	**0.81**	**0.82**	**0.93**
Vedic	0.82	**0.86**	0.76
Macro F1	54.4%	**63.2%**	46.3%

The most important confusion pattern (Figure 7) is the bidirectional Kavya–Shastra confusion: Kavya is predicted as Shastra by 9.7 texts on average, and Shastra is predicted as Kavya by 8.3 texts. The bidirectionality confirms that this is not a classifier bias artefact but a genuine morphological overlap. Prose Kavya texts (Harṣacarita, Daśakumāracarita, Kādambarī) share high nominalization and compound rates with Shastra prose, blurring the genre boundary. LDA performs worst ($F1 = 46.3\%$) and assigns zero recall to Puranic, reflecting its assumption of equal-covariance Gaussian distributions—violated by Shastra's enormous internal spread shown in Figure 4.

These results collectively establish that morphological features in PCA space carry genuine genre signal—63.2% macro F1 is well above the 20%

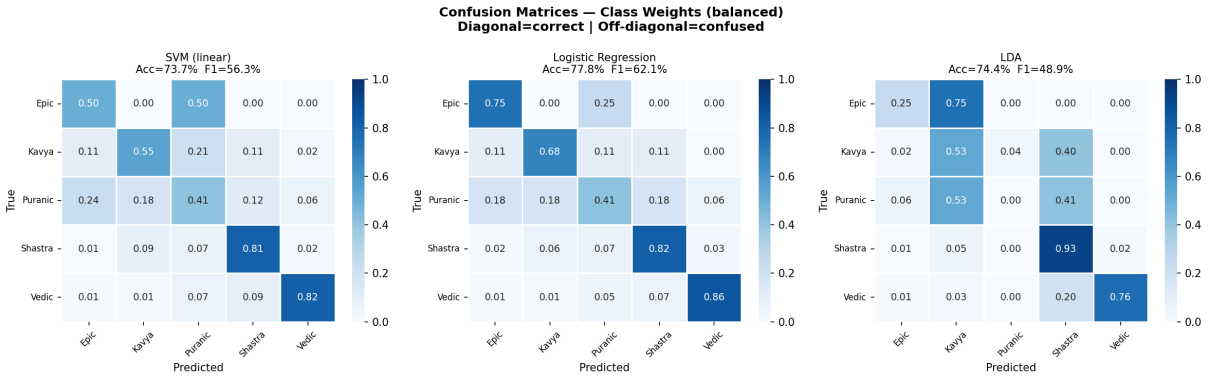

Figure 7: Normalized confusion matrices (rows = true genre). The primary error pattern—Kavya → Shastra (9.7 avg. texts) and Shastra → Kavya (8.3 texts)—is bidirectional, indicating genuine morphological similarity rather than classifier bias. Puranic recall is zero in LDA.

chance baseline—but that the Kavya–Shastra overlap and the small Epic sample impose a performance ceiling that no linear classifier can overcome without additional non-morphological features.

4.5 Diachronic Spearman correlations

All seven morphological features show highly significant Spearman correlations with text period (Table 4). The MCI trend ($r = -0.661^{***}$) and compound rate trend ($r = +0.634^{***}$) form the core diachronic finding: Sanskrit undergoes a systematic shift from morphological complexity toward nominal compounding across its 2,400-year history—a compensatory typological trade-off between inflection and compounding documented cross-linguistically(Fortson IV, 2010).

Table 4: Spearman correlations between morphological features and text period ($p < 0.001$ = ***, $p < 0.01$ = **, $p < 0.05$ = *).

Feature	r	Sig.
MCI	-0.661	***
Compound Rate	$+0.634$	***
NOUN Rate	$+0.589$	***
Mantra Rate	-0.568	***
Nom/Verb Ratio	$+0.525$	***
PRON Rate	-0.507	***
VERB Rate	-0.484	***

Within-genre correlations show that the diachronic trend is not merely a genre-level confound: MCI declines significantly within Kavya ($r = -0.459$, $p = 0.001^{**}$), Shastra ($r = -0.229$, $p = 0.010^{**}$), and Vedic ($r = -0.263$, $p = 0.022^{*}$) individually, demonstrating that the trend operates at both the corpus level and within individual genres.

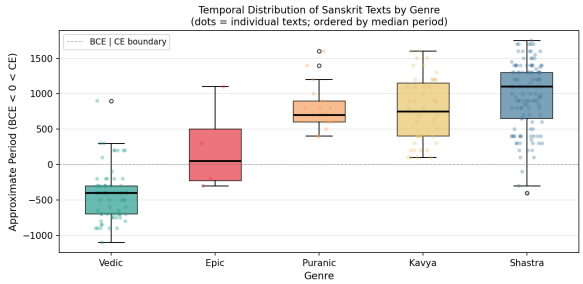

Figure 10: Temporal distribution of texts by genre. Vedic dominates the pre-CE period (median -450 BCE); Shastra is predominantly medieval (median $+1050$ CE). The 1,500-year temporal gap is a confound when comparing MCI across genres.

An important methodological caveat (Figure 10): Shastra's lower mean MCI is partly a temporal artefact—most Shastra texts are medieval (median $+1050$ CE), while Vedic texts are ancient (median -450 BCE). Genre and period are confounded in this corpus, and both register-driven and diachronic mechanisms contribute to the observed MCI differences.

Despite this confound, the within-genre diachronic trends confirm that the MCI decline is a genuine temporal signal and not purely a genre-composition artefact: Sanskrit grows morphologically simpler and nominally denser across time, regardless of which genre one examines.

5 Discussion

5.1 Hypothesis evaluation

Having established the empirical results, we now return to the four pre-registered hypotheses to assess which aspects of genre morphology are confirmed, which are falsified, and what the pattern of confirmation tells us.

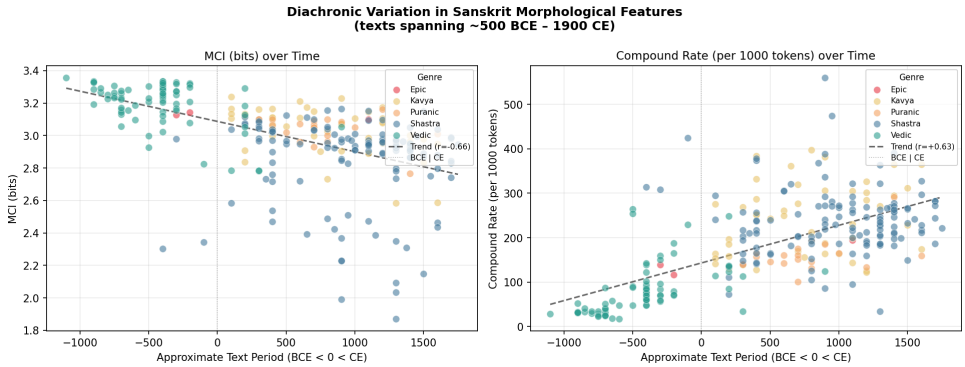

Figure 8: Diachronic variation in MCI and compound rate. MCI declines monotonically from Vedic texts ($\sim$1100 BCE) to late medieval Shastra ($\sim$1750 CE; $r = -0.66$, $p < 0.0001$). Compound rate increases correspondingly ($r = +0.63$, $p < 0.0001$), consistent with a compensatory inflection–compounding trade-off.

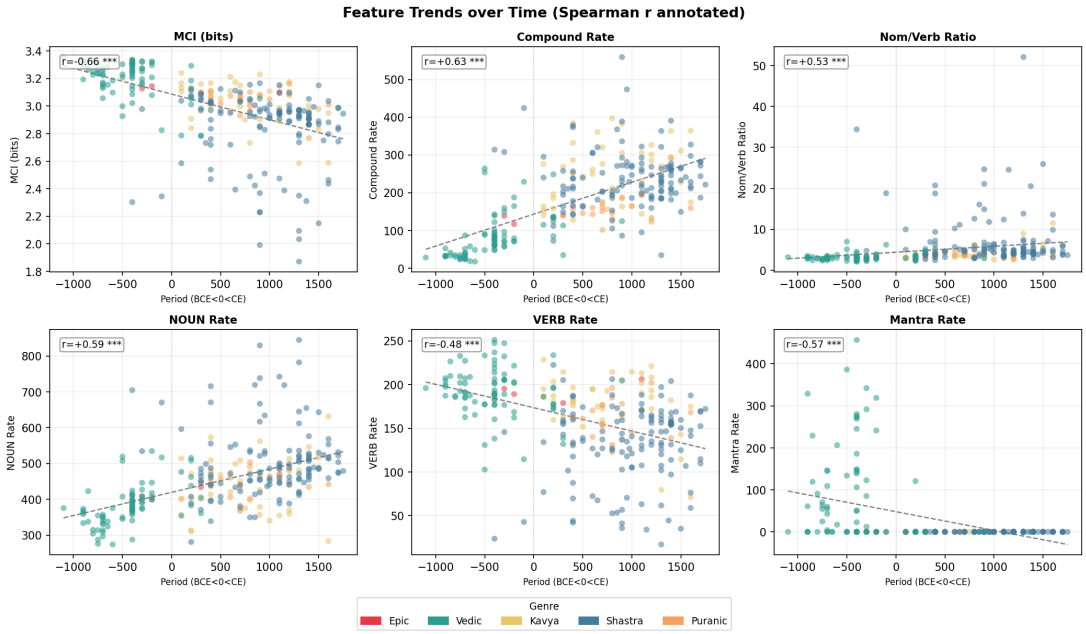

Figure 9: Feature trends over time (Spearman r annotated). All six features are significant at $p < 0.001$. NOUN rate and compound rate increase with period; VERB rate, pronoun rate, MCI, and mantra rate decrease.

H1 is *confirmed*: PC1 separates nominal (Shastra, Kavya) from verbal (Vedic) registers along a clear nominalization axis. The loading pattern—`pos_NOUN` and `compound_rate` positive, `pos_VERB` and `mood_Sub` negative—is exactly as predicted.

H2 is *not confirmed*: compound density is captured by PC1 (co-varying with nominalization), not PC2 as predicted. PC2 instead captures syntactic complexity (average dependency length) and adposition usage, driven partly by the Treebank annotation artefact.

H3 is *confirmed*: Vedic forms a 100% pure k-means cluster and separates at the root of the hierarchical dendrogram. Its distinctiveness arises from PC1 (verbal register), PC3 (optative mood), and the mantra rate feature, all reflecting archaic morphological features absent from Classical Sanskrit.

H4 is *partially disconfirmed*: Logistic Regression achieves the highest macro F1 (63.2%) with balanced class weights, outperforming SVM (54.4%). This reversal is consistent with LR's softer probabilistic optimization responding more flexibly to the reweighted loss landscape.

The pattern across H1–H4 reveals that the Vedic–Classical morphological divide (H1, H3) is strongly confirmed, while the finer within-Classical distinctions (H2, H4) are more ambiguous than predicted—suggesting that morphological features alone are sufficient to identify Vedic but not to fully resolve

Classical genre sub-categories.

5.2 The Kavya–Shastra boundary problem

The most linguistically significant finding is the bidirectional Kavya–Shastra confusion. Classical Sanskrit prose Kavya (Harṣacarita, Daśakumāracarita, Kādambarī) and technical Shastra prose share high compound density, high nominalization, and low verbal diversity. This morphological convergence is not a classification failure—it reflects a genuine historical fact: Sanskrit prose registers, regardless of their literary classification, converge toward the same nominal-compound-heavy style in the Classical period. The Kavya–Shastra distinction is ultimately a *content* distinction (aesthetic narrative vs. technical exposition) more than a *morphological* one, implying that genre disambiguation in this zone requires semantic or thematic features beyond what morphology alone can supply.

5.3 Genre as temporal confound

The genre period boxplot (Figure 10) reveals that Vedic (median -450 BCE) and Shastra (median $+1050$ CE) are separated by approximately 1,500 years on average. Shastra's lower MCI is therefore attributable to two partially confounded causes: (a) the register-specific nominalization that defines technical prose, and (b) the diachronic decline of morphological complexity affecting all later texts. Disentangling these explanations would require matched same-period comparison sets—a limitation of the current design that future work should address through diachronically-controlled regression models that partial out temporal effects before making genre-level morphological claims.

5.4 Limitations

1. Syntactic features (dependency length, verb-final rate) are only available for 1,766 of 15,900 chapters (Vedic Treebank), creating an artefactual PC2 sub-cluster within Vedic.
2. Epic has only 4 texts, providing insufficient statistical power for reliable per-genre conclusions; Epic results should be interpreted with particular caution.
3. Text period assignments are approximate (scholarly consensus midpoints); temporal uncertainty of ± 200 years is typical.
4. The mantra rate feature is zero for all non-Vedic texts (a clean discriminator) but also zero for some Vedic sutras, limiting its discriminative power within the Vedic category.

6 Conclusion

This paper has demonstrated that Sanskrit genre variation is a genuinely multidimensional morphological phenomenon, with four key findings:

1. PCA reveals a dominant nominalization axis (PC1, 30.3% variance) separating the verbal Vedic register from Classical Sanskrit genres, plus five additional axes capturing archaic verbal mood (PC3), syntactic word-order (PC4), and genre-internal variation.
2. Vedic is the only genre forming a coherent morphological cluster (100% k-means purity), while Kavya, Puranic, Epic, and Shastra are morphologically interleaved in a single overlapping region that unsupervised algorithms cannot resolve.
3. Balanced-weight Logistic Regression achieves 63.2% macro F1—a substantial improvement over the 20% chance baseline but below 70%—reflecting genuine morphological ambiguity at Classical genre boundaries that cannot be resolved by morphology alone.
4. Morphological complexity declines significantly across Sanskrit's 2,400-year history (MCI: $r = -0.661^{***}$) while compounding increases ($r = +0.634^{***}$), consistent with a cross-linguistic inflection–compounding compensatory trade-off(Fortson IV, 2010).

Future work should address the class imbalance in Epic (4 texts) through chapter-level feature extraction (the Mahabharata alone has 2,735 chapters), extend syntactic dependency features to non-Treebank chapters through automatic parsing, and apply diachronically-controlled regression to separate register and temporal effects on MCI.

At its core, this study establishes that two millennia of Sanskrit literary production are morphologically organized around a single dominant axis—the Vedic–Classical divide—and that within the Classical zone, genre is a matter of content and convention rather than of grammar.

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A Appendix

A.1 Qualitative Analysis of genre-boundary texts

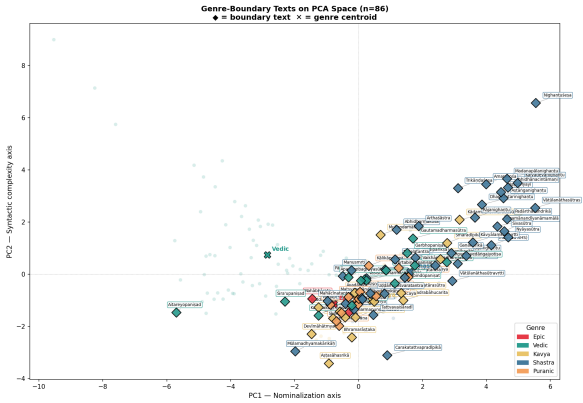

Figure 11: Genre-boundary texts (diamonds) on PCA space. Genre centroids (×) are shown for reference. The lexicographic Shastra texts (Nighaṇṭuśeṣa, Kaīyadevanighaṇṭu) occupy the extreme upper right; the most verbal Vedic text (Aitareyopaṇiṣad) appears at far left.

Eighty-six texts (32% of the corpus) were flagged as genre-boundary cases by misclassification (≥2/3 classifiers) or MCI outlier criteria. Qualitative analysis reveals three structural categories:

Category 1: Functionally hybrid texts. The Kādambarīsvīkaraṇasūtramajarī (Kavya, PC1=+3.16, MCI=2.59 bits) is a grammatical commentary on Kādambarī vocabulary. Despite Kavya classification, its feature profile—NOUN rate 631/1000, nom/verb ratio 11.5 ($z = +4.67$)—is indistinguishable from Shastra definitional prose. Similarly, the Garbhopaṇiṣat (Vedic, PC1=+1.52) is an Upanishad on embryology whose nominalized medical content (compound rate 229/1000, $z = +2.66$; NOUN rate 517/1000, $z = +2.33$) places it firmly in the Shastra region of morphological space. Haribhaktivilāsa (Kavya) exhibits extreme optative density (236/1000, $z = +4.55$) reflecting its ritual prescriptive register, closer to Dharmashastra than verse Kavya.

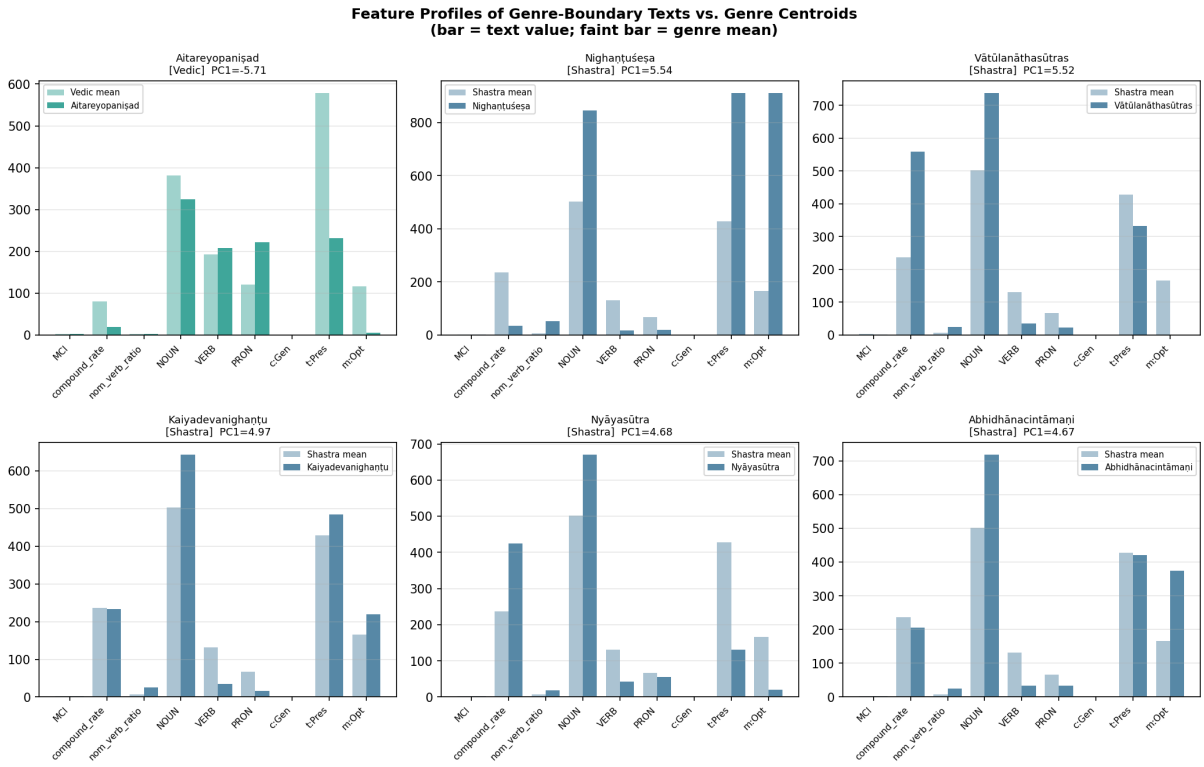

Figure 12: Feature profiles of the six most extreme boundary texts vs. their genre centroids. Nighaṇṭuśeṣa (NOUN rate 844/1000, MCI 1.87 bits) and the Vāṭūlanāthasūtras (compound rate 559/1000) exemplify opposite poles of the Shastra register.

Category 2: Register-extreme texts. Nighaṇṭuśeṣa (Shastra, MCI=1.87 bits, nom/verb ratio=52.1, $z = +6.66$) is a Sanskrit vocabulary list consisting almost entirely of nominative-singular nouns listed with synonyms. Its morphological entropy of 1.87 bits is the lowest in the entire corpus and defines the absolute minimum of inflectional diversity. Indu (ad AHS), a commentary on the Aṣṭāṅgahṛdayasaṃhitā, shows zero VERB rate, zero tense_Pres, zero mood_Ind—a text of nominal glosses with no finite verbs.

Category 3: Historically displaced texts. Twenty-five Vedic-labeled texts cluster morphologically with Shastra. The Ṛgvedavedāṅgajyotiṣa (Vedic, PC1=+2.76) is a Vedic astronomical text whose technical content (NOUN rate 518/1000, nom/verb ratio 7.0, $z = +4.61$; compound rate 254/1000, $z = +3.11$) is indistinguishable from Shastra. The Gautamadharmasūtra (Vedic, PC1=+1.70) is one of the earliest Dharmasutras; though Vedic by tradition, its terse nominalized legal prose (compound rate 264/1000, $z = +3.29$) matches Dharmashastra morphology. These texts reveal that the Vedic genre label encompasses a morphologically heterogeneous set of texts unified by religious tradition rather than grammatical register.